

Compact Vibrating Fork Level Switch For Solid Application Model: VFLS-400S



PRODUCT DESCRIPTION:



SOLID VIBRATING FORK – VFLS-400S is designed for reliable maximum and minimum level detection of free-flowing bulk solids in silos, hoppers and storage vessels. The vibrating fork principle provides dependable point-level detection of powders, granules and bulk materials. The operation is largely unaffected by variations in the material's conductivity and dielectric constant. The VFLS-400S is suitable for materials with a particle size of up to 15 mm, including plastic granules, silica, sand, cement, chalk, pulverized coal, PET, food grains, minerals, animal feed, sugar and wood chips. It is suitable for high-level and low-level detection, overfill protection, dry-run protection, level alarm and material feed control application.

OPERATING PRINCIPLE:

The VFLS-400S tuning fork is continuously vibrated at its mechanical resonant frequency by piezoelectric elements. When the process material covers the fork tines, the vibration is damped due to the resistance of the material. The change in vibration is electronically detected and processed by the internal electronics to generate a reliable level switching signal. When the material level falls below the fork, the tines resume their normal vibration and the switch returns to its normal operating state. The output relay provides a high- or low-level switching signal and can be used for level alarm and process control, such as controlling a feed motor, conveyor, pump, or evacuation valve.

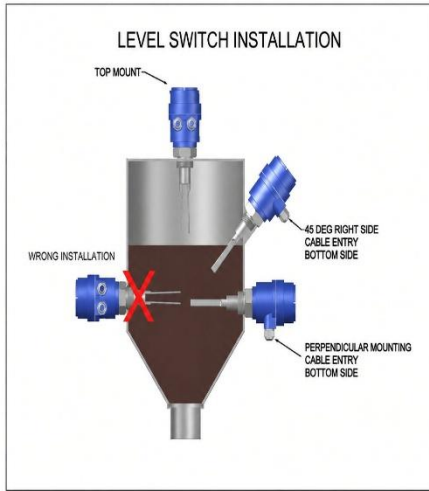
The VFLS-400S provides reliable point-level detection for various powders, granules and bulk solid materials, with no moving mechanical parts in the sensing mechanism.

SALIENT FEATURES:

- Compact design with integral electronics
- No calibration required – ready-to-use plug-and-play operation
- No moving parts – no mechanical wear and tear
- Universal power supply – suitable for AC/DC operation
- Short 100 mm fork design – minimum insertion depth for small silos and vessels
- Field-selectable High/Low Fail-Safe mode
- LED indication for normal, alarm and status conditions
- Adjustable switching delay for reliable operation and reduction of unwanted switching
- Suitable for powders, granules and free-flowing bulk solids
- Reliable high- and low-level detection
- Suitable for overfill and dry-run protection
- Suitable for silo, hopper, bin and process vessel applications
- Wide range of process materials and industrial applications

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PROBE CONSTRUCTION SUITABLE FOR MATERIAL BUILD-UP



The VFLS-400S is designed with a robust, compact and smooth tuning fork construction to minimize material accumulation on the sensing tines. The optimized fork geometry helps reduce the tendency of powder, granules and sticky bulk materials to accumulate between the tines, allowing the fork to maintain stable vibration for reliable level detection. The short and rigid fork design provides good mechanical strength and minimizes the insertion depth into the process vessel. The probe construction is suitable for applications where material coating and build-up may occur, while maintaining dependable High-Level and Low-Level switching performance.

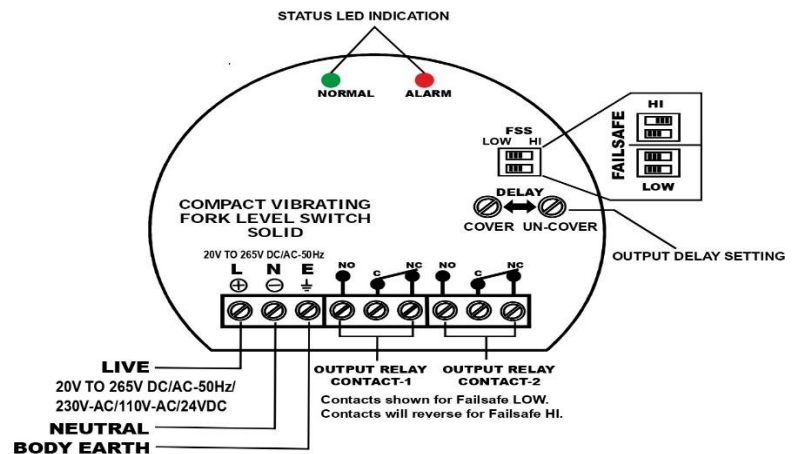
COMPONENTS OF COMPACT LEVEL SWITCH ARE EXPLAINED IN FIGURE



- ----Aluminum Housing
- ----Cable Entry
- -----Mounting Thread
- -----Extension pipe
- ----- Tuning Fork

ELECTRICAL CONNECTION UNIVERSAL /230 VAC/110VAC/24 VDC

ELECTRICAL CONNECTION- UNIVERSAL



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DATA SHEET

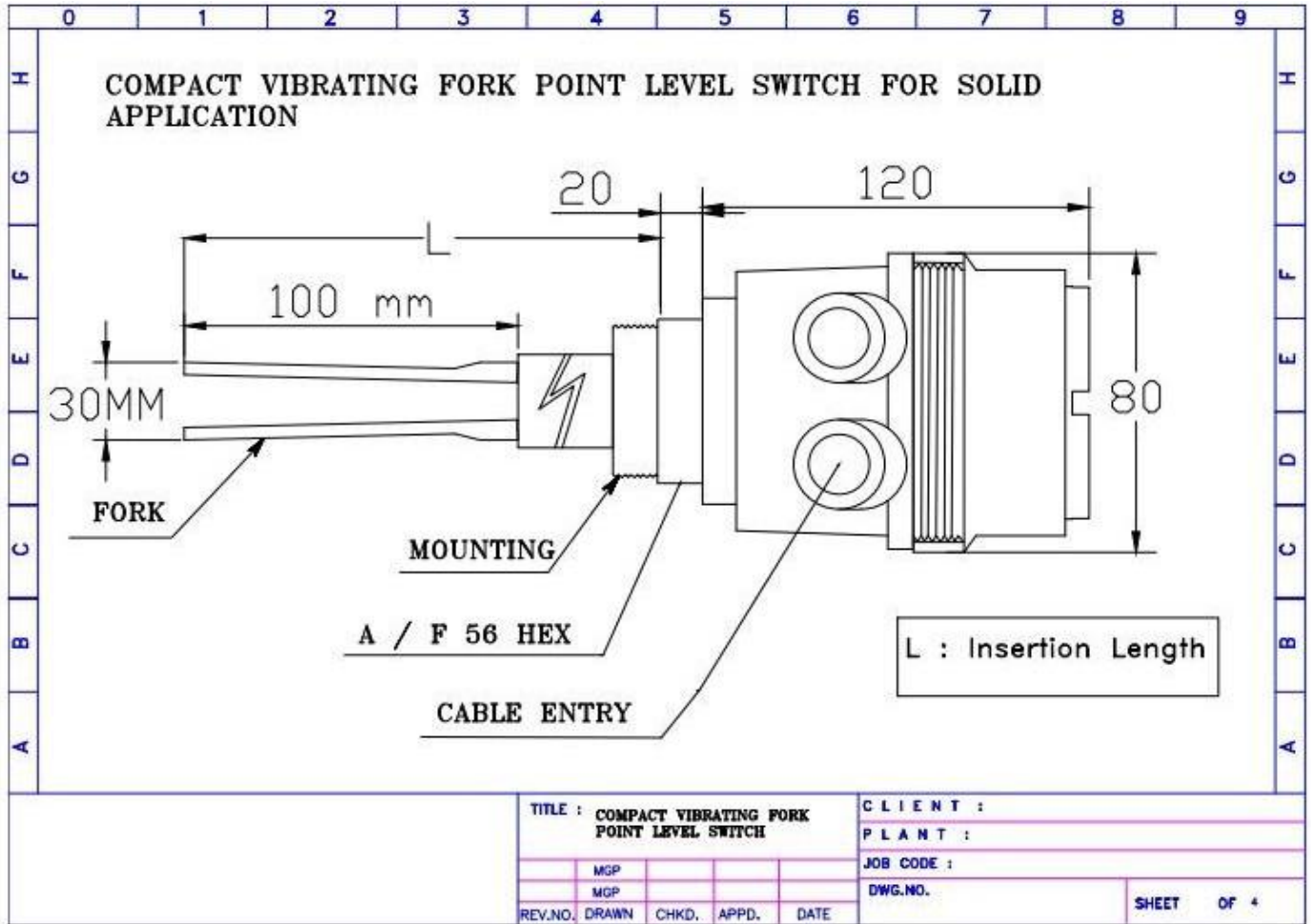
VFLS-400S Vibrating Fork Level Switch

Selection / Ordering Information

Selection	Code	Description
Model No.	VFLS-400	Vibrating Fork Level Switch
Application	S	Solids
Mounting Type	IN/RE	Integral / Remote
Probe Type	RD / RP	Rigid Pipe / Hose Flexible
Insulation Coating	NI/PT	Non- Coating /PTFE Coating
Probe Diameter	D1	Standard 30 mm
Probe Material	04 /4L/ 16/ 6L/MS/ SM	SS304/SS304L / SS316/ SS316L /Mild steel/ Special material
Power Supply	AC/DC/A1/UN	230VAC / 24VDC / 110VAC / Universal
Output	DP/SP/PN	DPDT Relay /SPDT Relay / PNP Relay
Switching Delay	ND / DA	No Delay / Adjustable
Indication	L	LED
Enclosure	WP / FL	Weatherproof / Flameproof
Connection	T	Threaded
	F	Flanged
Mounting	B1 / B2 / B3/B4	1" / 1½" / 2" BSP/ Special Mounting
	N1 / N2 / N3/N4	1" / 1½" / 2" NPT/ Special Mounting
Flange Rating	A1 / A3 / A6	ANSI 150# / ANSI 300# /ANSI 600#
Flange Facing	RF / FF	Raised Face / Flat Face
MOC-Process Connection	S4 / S6/ S16/MS/SL	SS304 / SS316 /SS316L /Mild steel / Special material
Flange Size	01 / 15 / 02 / 03 / 04/PS	1" / 1½" / 2" / 3" / 4"/ Special Process
Pressure	P1 /P2/P3/P4	Standard 0 to 2KG/ 2 To 10kg /10kg to 40kg / Customized
Temperature	T1	Max –150°C
Remote Cable	05 / SR	5 /Special Customer Requirement
Cable Gland	G1/ G2 / G3	PG13.5 / ½"NPT / ¾" ET
Material Certificate	M0 / MC	Not Required / Material test certificate
Test Certificate	TC / CC	Test Certificate / Calibration Certificate
Documentation	DS / GA / IR	Datasheet / GA / Inspection Report
Tagging	TAG	Customer Tag No.
Probe Length	XXXX	150 mm TO Customized

Compact Vibrating Fork Level Switch For Solid Application Model: VFSL-400S

Mechanical Drawing:



INFORMATION REQUIRED Before Purchase

Process Parameters: Process Connection, Process Temperature, Vessel pressure if any.

- Application media characteristics: Density, Viscosity, Sp. Gr.
- Supply Voltage, Probe length, Mounting etc.

Manufactured By:

Levtron Instruments Pvt Ltd.

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